# B1 Flows, Fluctuations and Complexity — Model Solutions, 2017

## Question 1 — Surface gravity–capillary waves

**Problem.** Derive Laplace's equation for the velocity potential of an incompressible irrotational flow. For deep-water linear surface waves with $\eta = A\sin(kx-\omega t)$ find $\phi$, the dispersion relation, the particle paths, and (with surface tension) $\omega^2 = c_1 k + c_2 k^3$.

### Laplace's equation

Irrotational: $\boldsymbol{\nabla}\times\mathbf{u} = 0$, so $\mathbf{u} = \boldsymbol{\nabla}\phi$.

$$\boldsymbol{\nabla}\cdot\mathbf{u} = \boldsymbol{\nabla}\cdot(\boldsymbol{\nabla}\phi) = \nabla^2\phi.$$

Incompressible: $\boldsymbol{\nabla}\cdot\mathbf{u} = 0$:

$$\boxed{\nabla^2\phi = 0.}$$

### Boundary conditions

**Kinematic** ($\partial_t\eta = \partial_z\phi$ at $z=0$): a fluid particle on the surface stays on the surface, so the vertical fluid velocity equals the surface vertical velocity (linearised: evaluated at $z=0$).

**Dynamic** ($\partial_t\phi = -g\eta$ at $z=0$): linearised Bernoulli at the free surface with $p = p_{\mathrm{atm}}$ gives $\partial_t\phi + g\eta + p_{\mathrm{atm}}/\rho = \mathrm{const}$; absorb the constant into $\phi$.

**Depth condition**: fluid at rest as $z\to-\infty$:

$$\boldsymbol{\nabla}\phi \to 0, \qquad \frac{\partial\phi}{\partial z}\to 0 \quad (z\to-\infty).$$

### Velocity potential and dispersion relation

Try separable form $\pi/2$ out of phase with $\eta$:

$$\phi(x,z,t) = f(z)\cos(kx-\omega t).$$

Insert in Laplace:

$$f''(z) - k^2 f(z) = 0.$$

General solution:

$$f(z) = Ce^{kz} + De^{-kz}.$$

Apply $z\to-\infty$ decay: $D=0$:

$$\phi = Ce^{kz}\cos(kx-\omega t).$$

Kinematic BC at $z=0$: $\partial_t\eta = \partial_z\phi$:

$$-A\omega\cos(kx-\omega t) = Ck\cos(kx-\omega t).$$

Solve for $C$:

$$C = -\frac{A\omega}{k}.$$

Velocity potential:

$$\boxed{\phi(x,z,t) = -\frac{A\omega}{k}\, e^{kz}\cos(kx-\omega t).} \tag{1}$$

Dynamic BC at $z=0$: $\partial_t\phi = -g\eta$:

$$-\frac{A\omega^2}{k}\sin(kx-\omega t) = -gA\sin(kx-\omega t).$$

Cancel and rearrange:

$$\boxed{\omega^2 = gk.} \tag{2}$$

## Particle paths

Lagrangian displacement $(\xi,\zeta)$ of a particle about $(x_0,z_0)$:

$$\dot{\xi} = u = \partial_x\phi, \qquad \dot{\zeta} = w = \partial_z\phi.$$

Evaluate at the mean position (linear theory) using $\phi = -(A\omega/k)e^{kz}\cos(kx-\omega t)$:

$$u = A\omega\, e^{kz_0}\sin(kx_0-\omega t), \qquad w = -A\omega\, e^{kz_0}\cos(kx_0-\omega t).$$

Integrate in time:

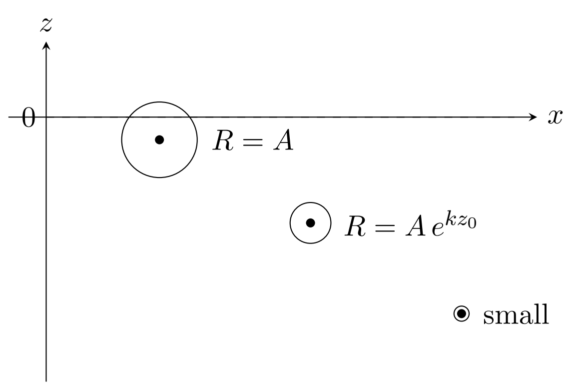
$$\xi = A\, e^{kz_0}\cos(kx_0-\omega t), \qquad \zeta = A\, e^{kz_0}\sin(kx_0-\omega t).$$

Eliminate $t$:

$$\xi^2+\zeta^2 = A^2e^{2kz_0}.$$

Circular paths of radius $R(z_0) = A\, e^{kz_0}$, decaying exponentially with depth:

$$\boxed{R(z_0) = A\, e^{kz_0}.}$$



### Surface tension: dispersion relation

Modified dynamic BC at $z = 0$:

$$\frac{\partial \phi}{\partial t} = -g\eta + \frac{\sigma}{\rho}\frac{\partial^2 \eta}{\partial x^2}.$$

Insert $\eta = A\sin(kx - \omega t)$, $\phi = -(A\omega/k)e^{kz}\cos(kx - \omega t)$:

$$-\frac{A\omega^2}{k}\sin = -gA\sin - \frac{\sigma}{\rho}k^2 A\sin\,.$$

Cancel $-A\sin$:

$$\frac{\omega^2}{k} = g + \frac{\sigma k^2}{\rho}.$$

Multiply by $k$:

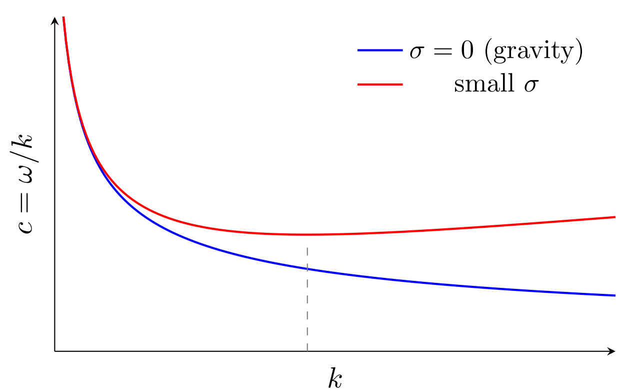
$$\boxed{\omega^2 = gk + \frac{\sigma}{\rho}k^3, \qquad c_1 = g, \quad c_2 = \sigma/\rho.} \tag{3}$$

Wave speed:

$$c(k) = \frac{\omega}{k} = \sqrt{\frac{g}{k} + \frac{\sigma k}{\rho}}.$$

Minimum at $k_* = \sqrt{g\rho/\sigma}$:

$$c_{\min} = \left(4g\sigma/\rho\right)^{1/4}.$$




Short waves ($k \gg k_*$, small $\lambda$) most affected: the curvature $\partial_x^2\eta \propto k^2$ amplifies the surface-tension restoring force, while gravity $g/k$ becomes negligible. Long waves ($k \ll k_*$) are gravity-dominated.

## Question 2 — Reynolds number, cylinder with source, Bernoulli, Torricelli

**Problem.** Derive Re by non-dimensionalising 2-D steady NS; for $\phi = Ur\cos\theta + \mu\cos\theta/r + m\log r$ around a cylinder of radius $a$ with outward normal velocity $V$, find $\mu, m$; for $V = 4U$ locate the upstream stagnation point; derive Bernoulli; find efflux speed and flux from a tank.

## Reynolds number

Steady 2-D incompressible NS:

$$(\mathbf{u}\cdot\boldsymbol{\nabla})\mathbf{u} = -\frac{1}{\rho}\boldsymbol{\nabla} p + \nu\nabla^2\mathbf{u}.$$

Scale: $\mathbf{u} = U\mathbf{u}^*$, $\boldsymbol{\nabla} = L^{-1}\boldsymbol{\nabla}^*$, $p = \rho U^2 p^*$:

$$\frac{U^2}{L}(\mathbf{u}^*\cdot\boldsymbol{\nabla}^*)\mathbf{u}^* = -\frac{U^2}{L}\boldsymbol{\nabla}^* p^* + \frac{\nu U}{L^2}\nabla^{*2}\mathbf{u}^*.$$

Divide by $U^2/L$:

$$(\mathbf{u}^*\cdot\boldsymbol{\nabla}^*)\mathbf{u}^* = -\boldsymbol{\nabla}^* p^* + \frac{1}{\mathrm{Re}}\nabla^{*2}\mathbf{u}^*.$$

Read off:

$$\boxed{\mathrm{Re} = \frac{UL}{\nu}.}$$

Behaviour: $\mathrm{Re} \ll 1$ Stokes (creeping) flow, viscosity dominates, time-reversible streamlines. $\mathrm{Re} \sim O(1)$ steady laminar flow with separation possible. $\mathrm{Re} \gtrsim 40$ unsteady wake (von Kármán street). $\mathrm{Re} \gtrsim 10^3$–$10^5$ transition to turbulence; inertia dominates, viscous effects confined to thin boundary layers.

## Cylinder with source: $\mu$, $m$

Velocity components from $\phi$:

$$u_r = \frac{\partial\phi}{\partial r} = U\cos\theta - \frac{\mu\cos\theta}{r^2} + \frac{m}{r},$$

$$u_\theta = \frac{1}{r}\frac{\partial\phi}{\partial\theta} = -U\sin\theta - \frac{\mu\sin\theta}{r^2}.$$

Apply normal BC $u_r(a,\theta) = V$ for all $\theta$:

$$U\cos\theta - \frac{\mu\cos\theta}{a^2} + \frac{m}{a} = V.$$

Match $\cos\theta$ part:

$$U - \frac{\mu}{a^2} = 0 \;\Rightarrow\; \mu = Ua^2.$$

Match constant part:

$$\frac{m}{a} = V \;\Rightarrow\; m = Va.$$

$$\boxed{\mu = Ua^2, \qquad m = Va.}$$

Potential and velocities:

$$\phi = U\cos\theta\Big(r + \frac{a^2}{r}\Big) + Va\,\log r,$$

$$u_r = U\cos\theta\Big(1 - \frac{a^2}{r^2}\Big) + \frac{Va}{r}, \qquad u_\theta = -U\sin\theta\Big(1 + \frac{a^2}{r^2}\Big).$$

## Stagnation point for $V = 4U$

By symmetry the stagnation point lies on $\theta = \pi$ (upstream axis). Set $u_\theta = 0$ automatically; require $u_r = 0$:

$$U \cos\pi\Big(1 - \frac{a^2}{r^2}\Big) + \frac{4Ua}{r} = 0.$$

Simplify $\cos\pi = -1$:

$$-U\Big(1 - \frac{a^2}{r^2}\Big) + \frac{4Ua}{r} = 0.$$

Cancel $U$, multiply by $r^2$:

$$-r^2 + a^2 + 4ar = 0.$$

Rearrange:

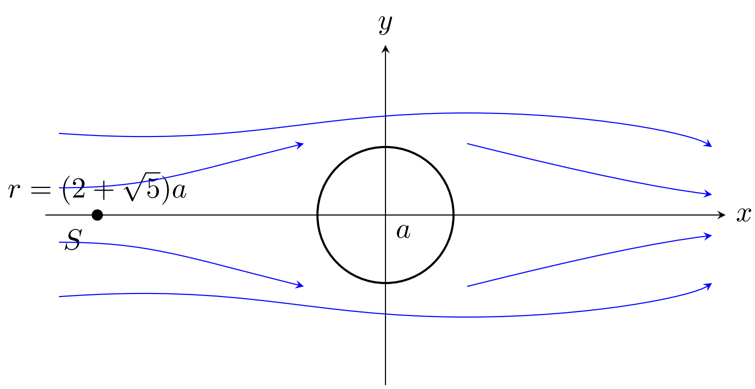
$$r^2 - 4ar - a^2 = 0.$$

Quadratic formula:

$$r = \frac{4a \pm \sqrt{16a^2 + 4a^2}}{2} = 2a \pm a\sqrt{5}.$$

Take $r > a$: $r = (2+\sqrt{5})a$. Distance from cylinder centre upstream:

$$\boxed{r_{\text{stag}} = (2+\sqrt{5})\,a.}$$




## Bernoulli along streamlines

Steady, inviscid, incompressible, gravity:

$$(\mathbf{u}\cdot\boldsymbol{\nabla})\mathbf{u} + \frac{1}{\rho}\boldsymbol{\nabla}p + g\hat{\mathbf{k}} = 0.$$

Use the identity:

$$(\mathbf{u}\cdot\boldsymbol{\nabla})\mathbf{u} = \boldsymbol{\nabla}\big(\tfrac{1}{2}u^2\big) + \boldsymbol{\omega}\times\mathbf{u}.$$

Substitute and multiply through by $\rho$:

$$\boldsymbol{\nabla}\big(\tfrac{1}{2}\rho u^2 + p + \rho g z\big) = -\rho\,\boldsymbol{\omega}\times\mathbf{u}.$$

Take dot product with $\mathbf{u}$:

$$\mathbf{u}\cdot\boldsymbol{\nabla}B = 0, \qquad B \equiv p + \tfrac{1}{2}\rho u^2 + \rho g z.$$

$(\mathbf{u}\cdot(\boldsymbol{\omega}\times\mathbf{u}) = 0.)$ Hence $B$ is constant along streamlines:

$$\boxed{p + \tfrac{1}{2}\rho u^2 + \rho g z = \text{const along streamlines.}}$$

### Tank: efflux speed and flux

Assumptions: steady, inviscid, incompressible, $A_e \ll A_0$ so surface speed $\approx 0$; both surface and exit at $p_{\text{atm}}$.

Bernoulli surface $\to$ exit:

$$p_{\text{atm}} + 0 + \rho g H = p_{\text{atm}} + \tfrac{1}{2}\rho u_e^2 + 0.$$

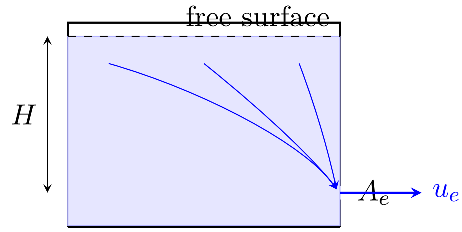
Solve:

$$u_e = \sqrt{2gH}.$$

Volume flux:

$$Q = A_e u_e.$$

$$\boxed{u_e = \sqrt{2gH}, \qquad Q = A_e\sqrt{2gH}.}$$




## Question 3 — Bifurcations and 2-D phase portrait

**Problem.** Analyse $\dot{x} = rx - 2x^2 + x^3$: fixed points, stability, bifurcations $r_1, r_2$, normal forms. Then phase-portrait $\dot{x} = x^2 - y - 1$, $\dot{y} = (x-2)y$.

### Fixed point $x^* = 0$

Set $\dot{x} = 0$:

$$x(r - 2x + x^2) = 0.$$

$x = 0$ is always a root. Stability from $f'(x) = r - 4x + 3x^2$:

$$f'(0) = r.$$

Stable for $r < 0$, unstable for $r > 0$. Bifurcation at:

$$\boxed{r_1 = 0.}$$

### Other fixed points

Roots of $x^2 - 2x + r = 0$:

$$x_\pm = 1 \pm \sqrt{1-r}.$$

Real for $r \le 1$. Stability:

$$f'(x_\pm) = r - 4x_\pm + 3x_\pm^2.$$

Use $x_\pm^2 = 2x_\pm - r$:

$$f'(x_\pm) = r - 4x_\pm + 3(2x_\pm - r) = 2x_\pm - 2r = 2(x_\pm - r).$$

Substitute $x_\pm$:

$$f'(x_\pm) = 2\big(1 - r \pm \sqrt{1-r}\big) = 2\sqrt{1-r}\big(\sqrt{1-r} \pm 1\big).$$

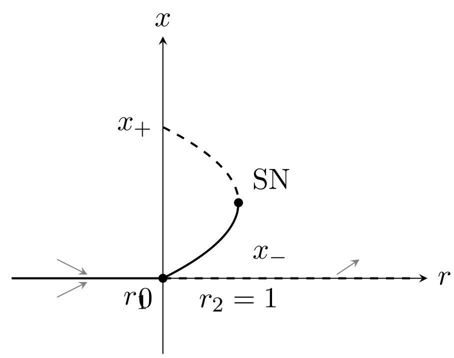
So $f'(x_+) > 0$ (unstable) and $f'(x_-) < 0$ for $0 < r < 1$ (stable); both vanish at $r = 1$.

Second bifurcation at:

$$\boxed{r_2 = 1.}$$

At $r = 1$, $x_\pm = 1$ collide (saddle-node). At $r = 0$, $x_- = 0$ collides with $x^* = 0$ (transcritical with $x_+ = 2$ existing for all $r \leq 1$).

### Bifurcation diagram and normal forms




Solid = stable, dashed = unstable.

**Normal form at $r_1 = 0$:** Taylor expand near $(x, r) = (0, 0)$:

$$\dot{x} = rx - 2x^2 + x^3 \approx rx - 2x^2.$$

Two fixed points 0 and $r/2$ exchange stability at $r = 0$:

$$\boxed{\dot{x} = rx - 2x^2 \quad \text{(transcritical).}}$$

**Normal form at $r_2 = 1$:** Let $r = 1 + \mu$, $x = 1 + y$:

$$\dot{y} = (1+\mu)(1+y) - 2(1+y)^2 + (1+y)^3.$$

Expand each term:

$$(1+\mu)(1+y) = 1 + y + \mu + \mu y,$$
$$-2(1+y)^2 = -2 - 4y - 2y^2,$$
$$(1+y)^3 = 1 + 3y + 3y^2 + y^3.$$

Sum:

$$\dot{y} = \mu + \mu y + y^2 + y^3.$$

Drop higher orders ($\mu y$, $y^3$) near $(0, 0)$:

$$\boxed{\dot{y} = \mu + y^2 \quad \text{(saddle-node).}}$$

## 2-D phase portrait

Fixed points: $\dot{y} = 0 \Rightarrow y = 0$ or $x = 2$.

*Branch $y = 0$:* $\dot{x} = x^2 - 1 = 0 \Rightarrow x = \pm 1$. Points $(1, 0)$, $(-1, 0)$.

*Branch $x = 2$:* $\dot{x} = 4 - y - 1 = 0 \Rightarrow y = 3$. Point $(2, 3)$.

Jacobian:

$$J = \begin{pmatrix} 2x & -1 \\ y & x-2 \end{pmatrix}.$$

At $(1, 0)$:

$$J = \begin{pmatrix} 2 & -1 \\ 0 & -1 \end{pmatrix}, \quad \tau = 1,\ \Delta = -2.$$

$\Delta < 0$: saddle. Eigenvalues $2, -1$; eigenvectors $(1, 0)^T$ (unstable), $(1, 3)^T$ (stable).

At $(-1, 0)$:

$$J = \begin{pmatrix} -2 & -1 \\ 0 & -3 \end{pmatrix}, \quad \tau = -5,\ \Delta = 6.$$

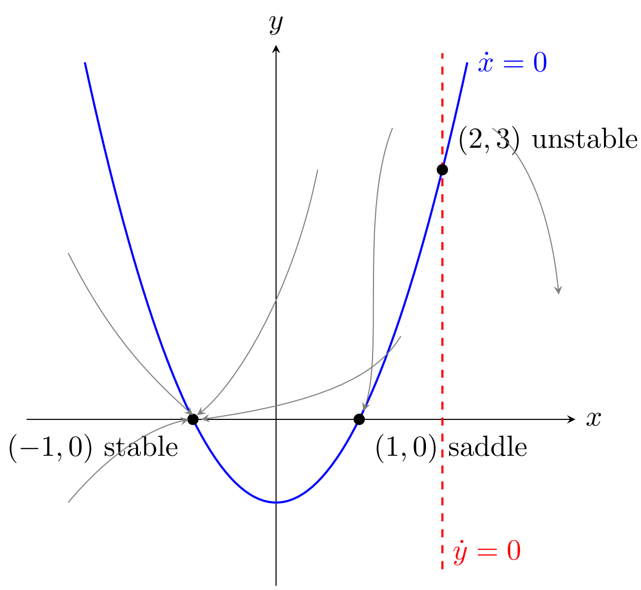
$\tau^2 - 4\Delta = 1 > 0$, $\tau < 0$, $\Delta > 0$: stable node. Eigenvalues $-2, -3$; eigenvectors $(1, 0)^T$ and $(1, 1)^T$.

At $(2, 3)$:

$$J = \begin{pmatrix} 4 & -1 \\ 3 & 0 \end{pmatrix}, \quad \tau = 4,\ \Delta = 3.$$

$\tau^2 - 4\Delta = 4 > 0$, $\tau > 0$, $\Delta > 0$: unstable node. Eigenvalues $1, 3$; eigenvectors $(1, 3)^T$ and $(1, 1)^T$.

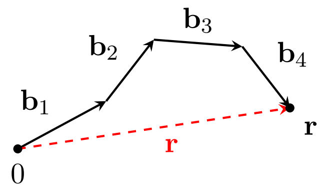
Nullclines: $\dot{x} = 0$ on parabola $y = x^2 - 1$; $\dot{y} = 0$ on $y = 0$ and $x = 2$.




# Question 4 — FJC, WLC and DNA

**Problem.** Show $\langle r^2 \rangle = Nb^2$ for FJC; derive WLC formula and limits; obtain Gaussian end-to-end distribution; show entropic spring $A = a + cr^2$ with $c$ in terms of $N, b, T$; high-force regime; numerical estimates for ds and ssDNA.

**FJC:** $\langle r^2 \rangle = Nb^2$




End-to-end vector:

$$\mathbf{r} = \sum_{i=1}^{N} \mathbf{b}_i, \qquad |\mathbf{b}_i| = b.$$

Square:

$$\mathbf{r} \cdot \mathbf{r} = \sum_{i,j} \mathbf{b}_i \cdot \mathbf{b}_j = \sum_i b^2 + \sum_{i \neq j} \mathbf{b}_i \cdot \mathbf{b}_j.$$

Take ensemble average; segments uncorrelated, $\langle \mathbf{b}_i \cdot \mathbf{b}_j \rangle = 0$ for $i \neq j$:

$$\boxed{\langle r^2 \rangle = Nb^2.}$$

## WLC: derivation and limits

Continuous tangent:

$$\mathbf{r} = \int_0^L \mathbf{t}(s)\,\mathrm{d}s, \qquad \langle \mathbf{t}(s) \cdot \mathbf{t}(s') \rangle = e^{-|s-s'|/P}.$$

Square and average:

$$\langle r^2 \rangle = \int_0^L \int_0^L e^{-|s-s'|/P}\,\mathrm{d}s\,\mathrm{d}s'.$$

Use symmetry $s' < s$, factor of 2:

$$\langle r^2 \rangle = 2 \int_0^L \mathrm{d}s \int_0^s e^{-(s-s')/P} \mathrm{d}s'.$$

Inner integral:

$$\int_0^s e^{-(s-s')/P} \mathrm{d}s' = P\big(1 - e^{-s/P}\big).$$

Outer integral:

$$\langle r^2 \rangle = 2P \int_0^L \big(1 - e^{-s/P}\big) \mathrm{d}s.$$

Evaluate:

$$\langle r^2 \rangle = 2P\big[L + P(e^{-L/P} - 1)\big].$$

$$\boxed{\langle r^2 \rangle = 2P^2 \left( e^{-L/P} - 1 + \frac{L}{P} \right).}$$

**Limit $L \gg P$:** $e^{-L/P} \to 0$:

$$\langle r^2 \rangle \to 2PL.$$

Identifies effective Kuhn length $b_K = 2P$ and segment count $L/(2P)$: Gaussian-coil regime.

**Limit $L \ll P$:** expand $e^{-L/P} = 1 - L/P + L^2/(2P^2) - \ldots$:

$$\langle r^2 \rangle \to L^2.$$

Rigid-rod regime; chain too short to bend.

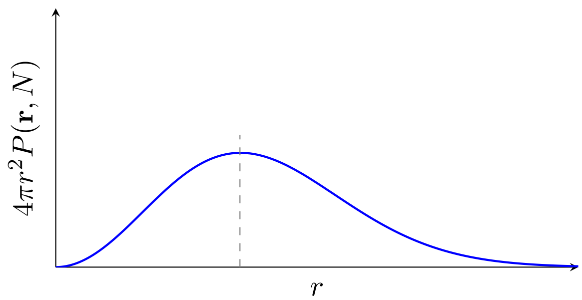
## End-to-end distribution and width

3-D Gaussian random walk: $x, y, z$ each diffuse independently in $N$ steps, with effective $2Dt \to b^2/3$ per step (since $\langle b_x^2 \rangle = b^2/3$). Sum of $N$ steps: variance $Nb^2/3$ per axis. Product of three Gaussians:

$$P(\mathbf{r}, N) = \left(\frac{3}{2\pi N b^2}\right)^{3/2} \exp\left(-\frac{3r^2}{2Nb^2}\right).$$


$$\boxed{P(\mathbf{r}, N) = \left(\frac{3}{2\pi N b^2}\right)^{3/2} \exp\left(-\frac{3r^2}{2Nb^2}\right).}$$


Width along one axis $\sigma = \sqrt{Nb^2/3}$. Total $\langle r^2 \rangle = 3\sigma^2 = Nb^2$, recovering the FJC result.




## Entropic spring

Boltzmann entropy of conformations with end-to-end vector $\mathbf{r}$:

$$S(\mathbf{r}) = k_B \ln \Omega(\mathbf{r}) = k_B \ln\big[V P(\mathbf{r}, N)\big].$$

Insert Gaussian:

$$S(\mathbf{r}) = k_B \ln C - \frac{3k_B r^2}{2Nb^2},$$

with $C$ collecting $r$-independent prefactors. At fixed $T$, with internal energy independent of $\mathbf{r}$:

$$A(\mathbf{r}) = -TS(\mathbf{r}) + \text{const.}$$

Combine:

$$\boxed{A(\mathbf{r}) = a + cr^2, \qquad c = \frac{3k_B T}{2Nb^2}.}$$

Quadratic in $r \Rightarrow$ Hookean restoring force $\mathbf{F} = -\boldsymbol{\nabla} A = -2c\mathbf{r}$. Stiffness comes purely from entropy (more conformations available at small $|\mathbf{r}|$): an *entropic spring*.

## High-force limit

Each segment aligns with applied force $\mathbf{F} = F\hat{\mathbf{z}}$; partition function for one segment:

$$Z_1 = \int \frac{\mathrm{d}\Omega}{4\pi}\, e^{\beta F b \cos\theta} = \frac{\sinh(\beta F b)}{\beta F b}.$$

Mean projection per segment:

$$\langle b_z \rangle = \frac{\partial \ln Z_1}{\partial(\beta F)} = b\,\mathcal{L}(\beta F b), \qquad \mathcal{L}(x) = \coth x - \tfrac{1}{x}.$$

Total extension:

$$z = Nb\,\mathcal{L}(\beta F b).$$

High-force ($\beta F b \gg 1$): $\coth(\beta F b) \to 1$, $1/(\beta F b) \to 0$:

$$z \to Nb\big(1 - \tfrac{k_B T}{Fb}\big).$$

Solve for $F$:

$$\boxed{F \simeq \frac{k_B T}{b}\,\frac{1}{1 - z/(Nb)}.}$$

$F$ diverges as the chain approaches contour length $L = Nb$.

## Bacterial DNA — numerical estimates

**Double-stranded:**

$$L = 5 \times 10^6 \times 0.34\,\text{nm} = 1.7 \times 10^6\,\text{nm} = 1.7\,\text{mm}.$$

$P = 150 \times 0.34\,\text{nm} = 51\,\text{nm}$. $L \gg P$:

$$\langle r^2 \rangle \approx 2PL = 2(51)(1.7 \times 10^6)\,\text{nm}^2 = 1.73 \times 10^8\,\text{nm}^2.$$

$$\boxed{\sqrt{\langle r^2 \rangle}_{\text{ds}} \approx 1.32 \times 10^4\,\text{nm} = 13.2\,\mu\text{m}.}$$

**Single-stranded:** $P_{\text{ss}} = 5 \times 0.34\,\text{nm} = 1.7\,\text{nm}$:

$$\langle r^2 \rangle \approx 2P_{\text{ss}}L = 2(1.7)(1.7 \times 10^6)\,\text{nm}^2 = 5.78 \times 10^6\,\text{nm}^2.$$

$$\boxed{\sqrt{\langle r^2 \rangle}_{\text{ss}} \approx 2.40 \times 10^3\,\text{nm} = 2.4\,\mu\text{m}.}$$

**Probability ratio** $P(R{=}0)/P(R{=}50\,\mu\text{m})$. Gaussian (with $\langle r^2 \rangle = Nb^2$, identifying $Nb^2 \to \langle r^2 \rangle$):

$$\frac{P(0)}{P(R)} = \exp\!\left(\frac{3R^2}{2\langle r^2 \rangle}\right).$$

ds: $R = 5 \times 10^4\,\text{nm}$, $\langle r^2 \rangle = 1.73 \times 10^8\,\text{nm}^2$:

$$\frac{3R^2}{2\langle r^2 \rangle} = \frac{3(2.5 \times 10^9)}{2(1.73 \times 10^8)} = 21.7.$$

$$\boxed{P(0)/P(50\,\mu\text{m})_{\text{ds}} \approx e^{21.7} \approx 2.7 \times 10^9.}$$

ss: $\langle r^2 \rangle = 5.78 \times 10^6\,\text{nm}^2$:

$$\frac{3R^2}{2\langle r^2 \rangle} = \frac{3(2.5 \times 10^9)}{2(5.78 \times 10^6)} = 649.$$

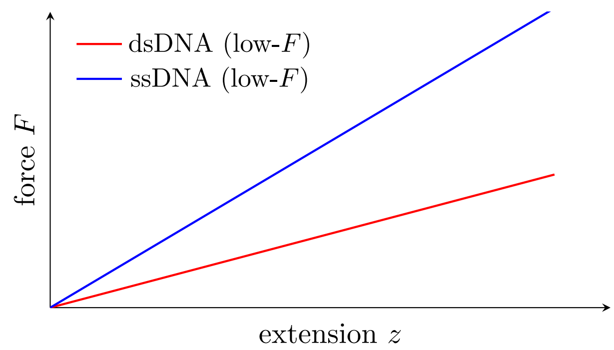
$$\boxed{P(0)/P(50\,\mu\text{m})_{\text{ss}} \approx e^{649} \sim 10^{282}.}$$

Comment: ssDNA is far more compact than dsDNA (smaller $P$, smaller $\langle r^2 \rangle$); reaching $50\,\mu$m is astronomically improbable. Both ratios $\gg 1$: extended states are heavily entropically suppressed.

**Low-force regime.** Both behave as Hookean entropic springs with stiffness $k = 3k_BT/\langle r^2 \rangle$:

$$F = \frac{3k_BT}{\langle r^2 \rangle}\,z.$$

ssDNA has smaller $\langle r^2 \rangle \Rightarrow$ steeper slope (stiffer at low force). dsDNA has larger $\langle r^2 \rangle \Rightarrow$ softer at low force.




ssDNA (steeper) is entropically stiffer because its smaller persistence length yields a more compact equilibrium coil and thus a larger $3k_BT/\langle r^2 \rangle$.